Mid-Air Word-Gesture Typing from a Single Uncalibrated Webcam

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1. Introduction

Text entry is the narrowest channel in most post-desktop computing systems. Whenever a device carries a camera but no keyboard — a mixed reality headset, a pair of lightweight augmented reality glasses, an information kiosk, a smart television, or a laptop being operated at a distance — the user must produce text without the mechanical constraints that make physical typing fast and accurate. Removing those constraints removes three things at once: the tactile confirmation that a key has been struck, the passive alignment that keeps the fingers registered to the layout, and the physical stops that bound articulation error. Every mid-air text entry system is, in effect, an attempt to replace these three functions with perception and inference. The resulting design problem has attracted sustained attention, and recent work has established mid-air typing as a viable, if still inefficient, modality [1, 2].

Within this space the *word-gesture keyboard* is the dominant high-performance technique. Rather than selecting one character at a time, the user traces a continuous path through the letters of a word and a probabilistic decoder

maps the entire trajectory onto a lexicon entry [3, 4]. The approach is attractive precisely because it does not require any individual point along the path to be accurate: the decoder scores the shape of the whole gesture against a dictionary, so localisation errors that would be fatal to per-key selection are absorbed by the matching process. Vulture transplanted this idea to mid-air, projecting hand movement onto a display and using a pinch as the word delimiter, and reported an entry rate of 20.6 words per minute (WPM), rising to 28.1 WPM once participants had trained on a set of phrases [5]. More recent systems have extended mid-air gesture typing with learned decoders that generalise across devices and keyboard geometries [6] and with dedicated input hardware [7].

A pattern runs through all of this work: *the sensing hardware is always privileged*. Vulture relied on a depth-based motion capture arrangement; ATK recovered ten-finger poses from a depth sensor in order to support freehand typing in air [8]; current extended reality systems inherit the multi-camera stereo rigs built into commercial headsets; and where such hardware is unavailable, researchers add instrumentation rather than accept degraded sensing. RingGesture is explicit on this point, introducing electrodes to delimit gesture trajectories and inertial units for hand tracking precisely because lightweight augmented reality glasses cannot accommodate enough cameras for reliable hand tracking, and reports an average of 27.3 WPM with a peak of 47.9 WPM [7].

The contrast with camera-only systems is stark. Work that restricts itself

to a single commodity webcam abandons the gesture paradigm altogether and reverts to per-key selection. A representative system of this kind combines an LSTM over MediaPipe Holistic landmarks with a thumb-to-index tap for key activation, reaching 98.52% gesture recognition accuracy and 97.86% character input accuracy, but an entry rate of only 37 characters per minute — roughly 7 WPM [9]. Between the depth-sensor gesture keyboards and the webcam per-key keyboards lies a three- to four-fold throughput gap. For context, a 2025 analysis of 176 XR text entry techniques reports a mean entry rate of 20.4 WPM across the 137 techniques that report WPM at all [1], placing the camera-only systems well below the field as a whole.

The gap in the literature. That gap has never been attributed. Because the sensing modality and the interaction technique have always varied together, it is not known whether mid-air word-gesture typing *requires* depth sensing, or whether depth has simply always been available to the researchers who studied it. The two candidate explanations for the observed difference in throughput — that monocular sensing is inadequate for gesture typing, or that camera-only systems have simply chosen a weaker interaction technique — predict the same published numbers and cannot be distinguished from the existing literature. This matters practically, because a monocular RGB camera is the sensor actually present on the widest class of deployed devices, and it matters methodologically, because the contribution of hardware to reported entry rates is currently unmeasurable.

Our proposal. We present WebSwipe, a bimanual mid-air word-gesture

keyboard that operates on a single, uncalibrated RGB webcam, with no depth sensor, no checkerboard calibration, no fiducial markers and no per-user enrolment step. The user is shown an on-screen keyboard overlay, pinches to begin a word, traces the shape of that word in the air, and releases the pinch to commit it; the non-dominant hand supplies space, deletion and selection from the candidate list. Two observations make this feasible on hardware that prior work has treated as inadequate.

First, when the hand moves in a plane approximately parallel to the image plane — which is the natural posture in front of a laptop screen — monocular projection preserves the *shape* of a gesture up to a similarity transform. A word-gesture decoder consumes normalised planar shape and discards absolute position, absolute scale and depth. The information a monocular camera fundamentally cannot recover is therefore, to a first approximation, the information the decoder discards anyway. This also implies that no camera calibration is needed, since neither the focal length nor the user's distance survives the normalisation step.

Second, gesture-level decoding averages sensor noise in a way that per-key selection cannot. A gesture is scored over many resampled trajectory points, so independent localisation errors contribute a term that is common to every candidate word and therefore does not, in expectation, reorder the candidate list. Per-key selection decides each character from a single observation, and errors compound multiplicatively across the characters of a word. The word-gesture paradigm is thus not merely compatible with a low-fidelity sensor;

it is specifically advantageous in that regime, which inverts the assumption implicit in the existing literature.

A third, practical observation supports the design. Pinch onset and pinch release both occur while the hand is nearly stationary, because the user pinches, then moves, then stops, then releases. The fast middle portion of a gesture, where motion blur is most severe and landmark quality is worst, requires no contact decision at all. The difficult problem of continuous contact detection under motion therefore reduces to two decisions taken under favourable conditions.

The salient contributions of the paper are,

1. A bimanual mid-air word-gesture keyboard operating from a single uncalibrated RGB webcam, without depth sensing, camera calibration, instrumented hardware or a per-user setup procedure. To our knowledge no prior word-gesture keyboard has been demonstrated under this sensing configuration.
2. A formal characterisation of why gesture-level decoding survives monocular landmark noise. We show that the shape-channel score is invariant to user–camera distance, to focal length and to image-plane translation, and that under isotropic landmark noise the score is offset by a term identical across all lexicon candidates, leaving the ranking unbiased with variance decaying as $O(1/N)$ in the number of resampled trajectory points. Per-key decoding admits no comparable averaging, which quantifies the advantage of the gesture paradigm specifically in

the low-fidelity-sensor regime.

3. A self-calibrating perception front end that eliminates per-user setup, comprising an online Gaussian-mixture pinch threshold, a running anthropometric scale normaliser and an interaction volume estimated implicitly from observed hand positions, evaluated for invariance across user–camera distance, illumination and hand appearance.

The organization of the paper is as follows. Section 2 presents a discussion of recent mid-air and extended reality text entry systems, of word-gesture decoding methods, of camera-only virtual keyboards and of the current state of monocular hand pose estimation, together with the limitations of the existing approaches. Section 3 presents the proposed WebSwipe architecture, the pinch state machine and decoding algorithms, and the formal analysis supporting the novelty claims. Section 4 discusses the dataset, the experimental setup, and a comparative analysis of the results with discussion. Section 5 presents the conclusion and opens further discussion with the future scope of the work.

References

[1] A. Bhatia, M. H. Mughrabi, D. Abdlkarim, M. Di Luca, M. Gonzalez-Franco, K. Ahuja, H. Seifi, Text entry for XR trove (TEXT): Collecting and analyzing techniques for text input in XR, in: Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems, ACM, 2025. doi:10.1145/3706598.3713382.

[2] J. Hu, J. J. Dudley, P. O. Kristensson, Seeing and touching the air: Unraveling eye-hand coordination in mid-air gesture typing for mixed reality, in: Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems, ACM, 2025, p. Article 1222. doi:10.1145/3706598.3713743.

[3] P.-O. Kristensson, S. Zhai, SHARK2: A large vocabulary shorthand writing system for pen-based computers, in: Proceedings of the 17th Annual ACM Symposium on User Interface Software and Technology (UIST '04), ACM, 2004, pp. 43–52.

[4] S. Zhai, P. O. Kristensson, The word-gesture keyboard: Reimagining keyboard interaction, Communications of the ACM 55 (9) (2012) 91–101.

[5] A. Markussen, M. R. Jakobsen, K. Hornbæk, Vulture: A mid-air word-gesture keyboard, in: Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '14), ACM, 2014, pp. 1073–1082. doi:10.1145/2556288.2556964.

[6] J. Shen, K. Khaldi, E. Zhou, H. B. Surale, A. Karlson, Gesture2Text: A generalizable decoder for word-gesture keyboards in XR through trajectory coarse discretization and pre-training, in: 2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), IEEE, 2024.

[7] J. Shen, R. Boldu, A. Kalla, M. Glueck, H. B. Surale, A. Karlson, RingGesture: A ring-based mid-air gesture typing system pow-

ered by a deep-learning word prediction framework, arXiv preprint arXiv:2410.18100.

[8] X. Yi, C. Yu, M. Zhang, S. Gao, K. Sun, Y. Shi, ATK: Enabling ten-finger freehand typing in air based on 3D hand tracking data, in: Proceedings of the 28th Annual ACM Symposium on User Interface Software and Technology (UIST '15), ACM, 2015, pp. 539–548. doi:10.1145/2807442.2807504.

[9] B. Mallik, M. A. Rahim, A. S. M. Miah, K. S. Yun, J. Shin, Virtual keyboard: A real-time hand gesture recognition-based character input system using LSTM and MediaPipe holistic, Computer Systems Science and Engineering 48 (2) (2024) 555–570. doi:10.32604/csse.2023.045981.